Mikhail Razakov

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Quant Developer / Research Engineer | C++ · Python

Research engineer with professional compiler/static-analysis experience and a strong C++/Python algorithms background. I build correctness-critical analyses and experimental infrastructure using exact/reference oracles, differential/randomized/metamorphic testing, reproducible harnesses, and conservative LLVM transformations.

Professional engineering

ISP RAS: industrial C#/.NET static analysis; MCST: LLVM 22, C++23, optimization legality, and verification harnesses.

Algorithms & research

PS-form: conservative solver, exact oracle, randomized/metamorphic validation; UniversalToolchain: controlled verifier experiment.

C++ / Python research tooling

LLVM, Python harnesses, graph algorithms, differential execution, reproducible counterexamples, and test oracles.

EXPERIENCE

2026 — present

ISP RAS — Static Analysis Engineer

- Contribute to SharpChecker, ISP RAS's industrial static-analysis platform for C#/.NET.

July 1 — August 31, 2026

MCST — Compiler Engineering Intern

- Implemented an LLVM LICM pass in C++; transformation legality is derived from loop structure, memory accesses, side effects, and speculative-safety constraints.
- Built a Python harness comparing execution before/after optimization and checking explicit transformation / no-transformation cases; also implemented RPO, Dijkstra, Dinic, and Tarjan SCC in C++23.

PROJECTS

PS-form Memory Dependence Analyzer

Conservative yes/no/maybe results checked by an independent exact oracle on small domains, randomized/metamorphic tests, and reproducible counterexamples.

LLVM Interprocedural Global IV Optimization

A conservative LLVM 22 prototype/MVP with an actual IR transform; 29 positive/negative regression cases cover verifier validity, idempotence, and before/after execution.

UniversalToolchain

Current exact test manifest: 1,324/1,324. In a frozen-corpus verifier experiment, B2 detected 32/32 primary and 10/10 challenge operators with 0/100 false positives on valid controls; four post-freeze holdouts were checked separately.

SKILLS

Programming
C++23, Python, C17, C#, Rust

Algorithms & analysis
data structures, graph algorithms, CFG/SSA, program analysis, memory dependence

Research engineering
exact/reference oracles, differential/randomized/metamorphic testing, reproducible harnesses, counterexamples

Systems
LLVM 22, x86-64, Linux, CMake, generated-code analysis

EDUCATION

HSE University — Software Engineering, 2026–2030

RECOGNITION

Prize-winner in HSE Vysshaya Proba in competitive and industrial programming.

Moscow / remote